

# Problem 4. Implementing Gradient Computation

$$f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = x_1 - 2x_2 + \frac{x_3}{4}, \quad g\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ x_2 \\ -x_1 + 3 \end{bmatrix}$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} : \text{input to } g$$

$$i) f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 1 & -2 & 1/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 0 = x_1 - 2x_2 + \frac{x_3}{4}$$

$$ii) g\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ x_2 \\ -x_1 + 3 \end{bmatrix}$$

$W_1 \quad (3, 2) \qquad \qquad \qquad b_1 \quad (3, 1) \qquad \qquad \qquad (3, 1)$

$$v) z_2 = f(z_1) = W_2 z_1 + b_2$$

$$z_2 = \begin{bmatrix} w_1^{(2)} & w_2^{(2)} & w_3^{(2)} \end{bmatrix} \begin{bmatrix} z_1^{(1)} \\ z_2^{(1)} \\ z_3^{(1)} \end{bmatrix} + b_2$$

$$\frac{\partial z_2}{\partial \text{vec}(w^{(2)})} = \begin{bmatrix} \frac{\partial z_2}{\partial w_1^{(2)}} & \frac{\partial z_2}{\partial w_2^{(2)}} & \frac{\partial z_2}{\partial w_3^{(2)}} \end{bmatrix} = \begin{bmatrix} z_1^{(1)} & z_2^{(1)} & z_3^{(1)} \end{bmatrix}$$

$$\frac{\partial z_2}{\partial b_2} = 1$$

$$vi) z_1 = g(x) = w_1 x + b_1$$

$$f(z_1) = w_2 z_1 + b_2$$

$$\frac{\delta(f \circ g)}{\delta z_1}(x) = w_2$$

$$vii) y = (f \circ g)(x), z_1 = g(x) = w_1 x + b_1$$

$$y = f(z_1) = w_2 z_1 + b_2$$

$$= w_2(w_1 x + b_1) + b_2$$

$$\frac{\delta y}{\delta z_1} = w_2^{(2)}$$

$$\boxed{\frac{\delta(f \circ g)}{\delta b_1} = \frac{\delta y}{\delta b_1} = \frac{\delta y}{\delta z_1} \frac{\delta z_1}{\delta b_1} = w_2^{(2)} I}$$

where  $I$  is the identity matrix

$$\frac{\delta(f \circ g)}{\delta \text{vec}(w^{(1)})} = \frac{\delta y}{\delta z_1} \frac{\delta z_1}{\delta \text{vec}(w^{(1)})} = w_2 \frac{\delta z_1}{\delta \text{vec}(w^{(1)})}$$

calculating  $\frac{\delta z^{(1)}}{\delta \text{vec}(w^{(1)})}$

$$\begin{pmatrix} z_1^{(1)} \\ z_2^{(1)} \\ z_3^{(1)} \end{pmatrix} = \begin{bmatrix} w_1^{(1)} & w_2^{(1)} \\ w_3^{(1)} & w_4^{(1)} \\ w_5^{(1)} & w_6^{(1)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ b_3^{(1)} \end{bmatrix}$$

$$\frac{\delta z^{(1)}}{\delta \text{vec}(w^{(1)})} = \begin{bmatrix} \frac{\delta z_1^{(1)}}{\delta w_1^{(1)}} & \frac{\delta z_1^{(1)}}{\delta w_2^{(1)}} & \frac{\delta z_1^{(1)}}{\delta w_3^{(1)}} & \frac{\delta z_1^{(1)}}{\delta w_4^{(1)}} & \frac{\delta z_1^{(1)}}{\delta w_5^{(1)}} & \frac{\delta z_1^{(1)}}{\delta w_6^{(1)}} \\ \frac{\delta z_2^{(1)}}{\delta w_1^{(1)}} & \frac{\delta z_2^{(1)}}{\delta w_2^{(1)}} & \frac{\delta z_2^{(1)}}{\delta w_3^{(1)}} & \frac{\delta z_2^{(1)}}{\delta w_4^{(1)}} & \frac{\delta z_2^{(1)}}{\delta w_5^{(1)}} & \frac{\delta z_2^{(1)}}{\delta w_6^{(1)}} \\ \frac{\delta z_3^{(1)}}{\delta w_1^{(1)}} & \frac{\delta z_3^{(1)}}{\delta w_2^{(1)}} & \frac{\delta z_3^{(1)}}{\delta w_3^{(1)}} & \frac{\delta z_3^{(1)}}{\delta w_4^{(1)}} & \frac{\delta z_3^{(1)}}{\delta w_5^{(1)}} & \frac{\delta z_3^{(1)}}{\delta w_6^{(1)}} \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & x_1 & x_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & x_1 & x_2 \end{bmatrix}$$

$$\text{unvec} \left[ \frac{\delta(f \circ g) x}{\delta \text{vec}(w^{(1)})} \right] = \begin{bmatrix} w_1^{(2)} & w_2^{(2)} & w_3^{(2)} \end{bmatrix} \begin{bmatrix} x_1 & x_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & x_1 & x_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & x_1 & x_2 \end{bmatrix}$$

$$= \begin{bmatrix} \omega_1^{(2)} x_1 & \omega_1^{(2)} x_2 & \omega_2^{(2)} x_1 & \omega_2^{(2)} x_2 & \omega_3^{(2)} x_1 & \omega_3^{(2)} x_2 \end{bmatrix}$$

$$= \begin{bmatrix} \omega_1^{(2)} x_1 & \omega_1^{(2)} x_2 \\ \omega_2^{(2)} x_1 & \omega_2^{(2)} x_2 \\ \omega_3^{(2)} x_1 & \omega_3^{(2)} x_2 \end{bmatrix} = \omega^{(2)T} x^T$$

$$\Rightarrow \boxed{\text{unvec} \left[ \frac{\delta(f \circ g) x}{\delta \text{vec}(\omega^{(n)})} \right] = \omega^{(2)T} x^T}$$